



SOFTWARE DEV

MERN STACK DEVELOPMENT REPORT

Task Manager App

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Lab Report

Project: Task Manager App (React Native / Expo) **Date:** August 15, 2026

Objective

Extend the existing task manager app with new features (folders, mind mapping, upcoming task view, progress tracking, ad monetization scaffold) and resolve related build/runtime issues, in preparation for Play Store submission.

Work Completed

1. Notes — Folder System

- Added folder creation, editing, deletion, and note-to-folder assignment
- Notes without a folder remain in a general list; deleting a folder moves its notes back to general rather than deleting them

2. Notes — Mind Map Feature

- Built an interactive mind map editor: node add/edit/delete, curved SVG connectors, pinch-to-zoom and pan, keyboard-aware toolbar, autosave, and manual save
- Fixed a node/connector alignment bug and a keyboard-overlap UX issue
- Fixed a crash caused by SVG gradient components under the app's New Architecture (Fabric) configuration; replaced with a solid-color progress ring approach

3. Bug Fixes

- Fixed an incorrect navigation route ('Home' → 'MainTabs') in the permission setup flow that caused a repeating navigation error
- Removed an unused dependency (`lodash`) and enabled Android code/resource shrinking to reduce app size

4. Home Screen

- Replaced the "Tomorrow" tab with an "Upcoming" tab: tasks are grouped by date and sorted by priority within each day
- Added a completion-progress indicator to the "Today" tab

5. Ad Monetization Scaffold

- Built a mock ad service (`adsService.js`) with an online/offline check and a rewarded-ad simulation, designed so real AdMob integration later requires changes to a single file
- Gated mind map creation behind a per-use rewarded-ad check (fails gracefully offline)
- Added an occasional full-screen ad overlay, shown app-wide with a cooldown period; a persistent banner approach was built and then removed in favor of a less intrusive UX

6. Build & Deployment Guidance

- Clarified native vs. JS-only changes and when a new EAS build is required
- Corrected build configuration for locally installable APKs vs. Play Store app bundles
- Outlined the Play Console closed-testing requirement (12 testers, 14 days) before production access

Files Added/Modified

Services: `adsService.js`, `foldersService.js` **Components:** `FolderCard.js`, `NoteCard.js`, `TodayProgressCard.js`, `UpcomingTaskList.js`, `FullScreenAdOverlay.js` **Screens:** `NotesScreen.js`, `FolderNotesScreen.js`, `AddEditNoteScreen.js`, `MindMapScreen.js`, `HomeScreen.js`, `PermissionSetupScreen.js` **Other:** `NoteContext.js`, `dateHelpers.js`, `mindMapLayout.js`, `AppNavigator.js`, `App.js`, `app.json`, `eas.json`, `package.json`

Status

Core features functional in local testing. App pending a production build and Play Console closed-testing cycle before public release.